



Enterprise VM Migration Platform

Production-grade toolkit for migrating VMware, Hyper-V, and Azure workloads to KVM/libvirt and Kubernetes — with zero downtime and full automation.

Apache 2.0 Open Source — Built for Enterprise Linux

The Problem

Why enterprises are leaving VMware — and what's at stake.

Broadcom Licensing Shock

VMware license costs increased 2-12x after the Broadcom acquisition. Perpetual licenses eliminated. Bundled pricing forces customers to pay for unused products.

Vendor Lock-in Risk

VMDK, vSphere APIs, and proprietary tooling create deep dependencies. Every year of delay increases switching costs exponentially.

End-of-Support Pressure

Legacy vSphere versions losing support. Security patches stop. Compliance audits flag unpatched hypervisors. Migration becomes urgent, not optional.

Cloud-Native Gap

VMware workloads can't natively run in Kubernetes. Modernization requires re-platforming VMs to KubeVirt — but manual migration is slow and error-prone.

"Organizations face a stark choice: absorb 300-1200% cost increases, or migrate to open alternatives. The window for planned migration is closing."

Cost Impact

The financial case for migration is overwhelming.

\$0

KVM/libvirt License Cost
(vs \$5,000+/socket/yr
VMware)

70%

Average TCO Reduction
over 3 years

10x

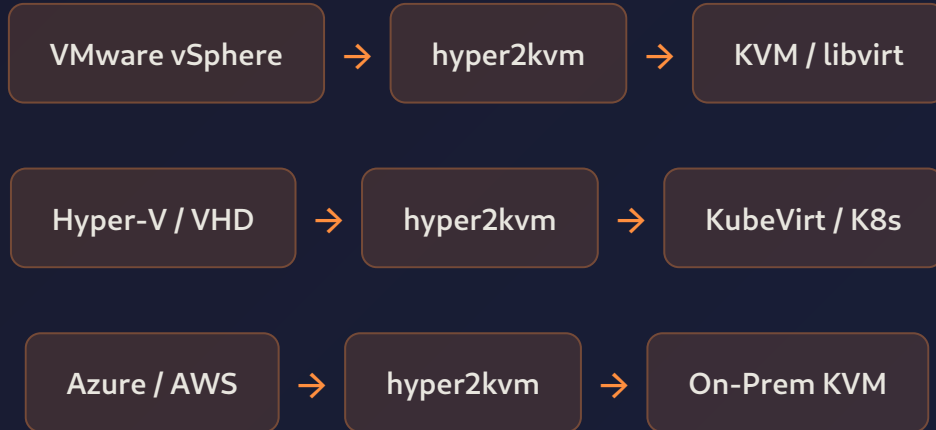
ROI on Migration
Investment

3-Year TCO Comparison (100 VMs)

Cost Category	VMware vSphere	KVM + KubeVirt	Savings
Hypervisor Licensing	\$500,000	\$0	\$500,000
Management Platform	\$200,000	\$0 (Cockpit/CLI)	\$200,000
Support Subscription	\$150,000	\$50,000 (RHEL optional)	\$100,000
Migration Tooling	N/A	\$0 (hyper2kvm)	\$0
Staff Training	\$30,000	\$20,000	\$10,000
Total	\$880,000	\$70,000	\$810,000

The Solution

hyper2kvm — one tool for every migration path.



Any Source Format

VMDK, OVA, OVF, VHD/VHDX, RAW, AMI — auto-detected. No manual format juggling.

Offline Guest Fixes

Automatic fstab repair, initramfs rebuild, driver injection, GRUB fix — VMs boot on first try.

Dual Deploy Target

Deploy to libvirt (traditional) or KubeVirt (cloud-native) from the same pipeline. Or both.

Terminal UI (zkvm)

Interactive TUI with vSphere discovery, VM selection, live progress, file browser — no CLI memorization needed.

Platform Capabilities

Enterprise features built for production workloads.

Capability	hyper2kvm	virt-v2v	Manual
VMware vSphere integration (govc/pyvmomi)	✓	✓	✗
Hyper-V / VHD support	✓	✓	✗
Azure VM export	✓	✗	✗
KubeVirt / Kubernetes deploy	✓	✗	✗
Interactive TUI (guided wizard)	✓	✗	✗
Windows driver injection	✓	✓	✗
Automatic fstab/GRUB repair	✓	Partial	✗
Live progress streaming	✓	✗	✗
Batch / queue migration	✓	✗	✗
YAML-driven automation	✓	✗	✗
Pre-flight validation	✓	✗	✗
Boot verification test	✓	✗	✗
Apache 2.0 open source	✓	✓ (GPL)	N/A

Migration **Workflow**

From VMware to production KVM in 4 steps.

1

Discover & Select

Connect to vCenter, discover all VMs with CPU/RAM/power state. Select targets interactively or via YAML. Pre-flight checks validate the environment.

2

Export & Convert

Export VMs via govc/OVF/HTTPS with live progress. Convert VMDK→qcow2 with compression and flattening. Supports multi-disk VMs and snapshots.

3

Fix & Optimize

Offline guest repairs: fstab stabilization (UUID/path), initramfs rebuild with virtio drivers, VMware tools removal, SELinux relabel, GRUB reconfiguration.

4

Deploy & Verify

Generate libvirt domain XML, register with virsh, or deploy to KubeVirt on Kubernetes. Automatic boot test validates the migrated VM starts correctly.

Minutes, Not Days

Typical VM migration: 5-30 minutes depending on disk size. No manual intervention required for standard Linux/Windows guests.

Batch Processing

Queue multiple VMs for sequential migration. Track progress per-VM. Failed VMs don't block the queue — results summarized at completion.

Why KubeVirt?

Run VMs and containers on the same platform — the best of both worlds.

Unified Platform

VMs and containers managed by Kubernetes. One API, one toolchain, one team.

Cloud-Native Ops

GitOps, CI/CD, Helm charts, operators — apply modern practices to VM workloads.

Gradual Modernization

Migrate VMs first, containerize later. No big-bang rewrite. Incremental path to cloud-native.

No Hypervisor Tax

KubeVirt runs on standard Linux + KVM. No per-socket licensing. Scale with commodity hardware.

Multi-Cloud Ready

Same KubeVirt manifests work on-prem, in AWS (EKS), Azure (AKS), or GCP (GKE). True portability.

Enterprise Backed

KubeVirt is a CNCF project, backed by Red Hat (OpenShift Virtualization). Production-proven at scale.

"KubeVirt lets you run your legacy VMs in Kubernetes today, and modernize them into containers tomorrow — on your schedule, not your vendor's."

Supported Platforms

Comprehensive source and target compatibility.

Migration Sources

- VMware vSphere 6.x / 7.x / 8.x
- VMware ESXi (standalone)
- Microsoft Hyper-V
- Microsoft Azure VMs
- AWS EC2 (AMI export)
- Raw disk images
- Any VMDK/OVA/OVF/VHD file

Deploy Targets

- KVM / libvirt (RHEL, Fedora, Ubuntu)
- KubeVirt on Kubernetes
- K3s + KubeVirt (edge)
- OpenShift Virtualization
- Proxmox VE (qcow2 import)
- Any KVM-based hypervisor

Guest OS Support

- RHEL 7 / 8 / 9 / 10
- CentOS, AlmaLinux, Rocky
- Ubuntu 18.04–24.04
- Debian 10–13
- SUSE / openSUSE
- Windows Server 2016–2025
- Windows 10 / 11
- Photon OS, Fedora, Arch

Output Formats

- qcow2 (default, compressed)
- RAW (maximum performance)
- VDI (VirtualBox compatible)
- Libvirt domain XML
- KubeVirt VirtualMachine YAML

Customer **Scenarios**

Real-world migration patterns hyper2kvm solves.

VMware Exit — Enterprise

Scenario: 500 VMs on vSphere, facing 5x license increase.

Solution: Batch migrate to RHEL + KVM using hyper2kvm YAML automation. Queue processing handles overnight migrations. Boot verification catches issues automatically.

Result: \$2M+ annual savings, zero vendor lock-in.

Cloud Repatriation

Scenario: Azure VMs costing \$50K/month, moving to on-prem.

Solution: Export Azure VMs via hyper2kvm's Azure integration. Convert VHD→qcow2, deploy to local KVM cluster.

Result: 60% cloud cost reduction, data sovereignty maintained.

Kubernetes Modernization

Scenario: Legacy VMs that can't be containerized yet, but team wants Kubernetes.

Solution: Migrate VMs to KubeVirt. Run alongside containers. Modernize incrementally.

Result: Unified platform, single operations team, gradual containerization path.

Edge / Factory Floor

Scenario: Industrial VMs running on expensive VMware licenses at remote sites.

Solution: K3s + KubeVirt on lightweight hardware. hyper2kvm converts and deploys.

Result: Zero license cost at edge, GitOps management from central cluster.

Why **hyper2kvm**?

What sets us apart from alternatives.

First-Boot Success

Offline guest fixes ensure VMs boot on first try. No manual driver installation, no broken fstab, no missing initramfs. We fix it before you notice.

Production-Grade

1,273 unit tests. Pre-flight validation. Structured logging with tracing. Error diagnosis. This isn't a script — it's a platform.

Interactive TUI

The zkvm terminal UI provides guided migration with live progress, file browser, vSphere discovery, and deploy toggles. No CLI memorization needed.

YAML Automation

Every migration is reproducible via YAML config. Perfect for CI/CD pipelines, GitOps workflows, and batch processing. Automate hundreds of VMs.

Ready to Eliminate VMware Costs?

hyper2kvm is open source (Apache 2.0) and ready for production. Start migrating today — no license required, no vendor approval needed.

Getting Started

From zero to migrated VM in 2 commands.

Quick Install

```
# Install
sudo ./scripts/quickstart.sh

# Migrate a VMDK
sudo h2kvmctl --cmd local --vmdk disk.vmdk --output-dir ./out

# Or use the interactive TUI
./zkvm/zkvm

# Or use YAML automation
sudo h2kvmctl --config migration.yaml
```

Example YAML Config

```
cmd: vsphere
vcenter: 10.0.0.100
vc_user: administrator@vsphere.local
vc_password_env: VC_PASSWORD
dc_name: Datacenter1
vs_action: export_vm
vs_vm: production-web-01
output_dir: ./out
out_format: qcow2
emit_domain_xml: true
regen_initramfs: true
deploy_k8s: true
```

```
k8s_vm_name: web-01
```

```
k8s_namespace: production
```

1,273

Unit Tests Passing

Apache 2.0

Open Source License

Python + Go

Dual Language Platform

hyper2kvm — Production-Grade Hypervisor to KVM Migration Toolkit

Built for the Enterprise Linux ecosystem (Fedora / RHEL / CentOS)

github.com/ssahani/hyper2kvm